$$\frac{d I_i}{d t} = \beta_i(t)\left(1 - \frac{I_i}{N_i}\right)\sum_j C_{ij}(t) I_j - \gamma_i(t) I_i \tag{1}$$

$$R_d(t) = \rho\left(K(t)\right), K(t) = diag\left(\beta_i(t)\left(1 - x_i(t)\right)\right) C(t) \Gamma(t)^{-1} \tag{2}$$